

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: `udp port 53 unreachable`

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: ICMP 203.0.113.2 line is the start of the error message indicating that the UDP packet was undeliverable to port 53 of the DNS server.

The port noted in the error message is used for: Port 53 is a port for DNS service and used for accessing the www.yummyrecipesforme.com website.

The most likely issue is: A problem with the server or a malicious attack

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 1:24 p.m.

Explain how the IT team became aware of the incident: Several customers reported that they were unable to access the website www.yummyrecipesforme.com, and saw the error "destination port unreachable" after waiting for the page to load.

Explain the actions taken by the IT department to investigate the incident: I attempted to visit the website and you also received the error "destination port unreachable." To troubleshoot the issue, you load your network analyzer tool, tcpdump, and attempt to load the webpage again. To load the webpage, your browser sends a query to a DNS server via the UDP protocol to retrieve the IP address for the website's domain name; this is part of the DNS protocol. Your browser then used this IP address as the destination IP for sending an HTTPS request to the web server to display the webpage. The analyzer shows that when you send UDP packets to the DNS server, you receive ICMP packets

containing the error message: “udp port 53 unreachable.”

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): A domain name to an IP address was never completed after the DNS request is sent.

Note a likely cause of the incident: DNS server problem or a form of DDOS/DOSS attack